

Sensitivity of GLOBAL1D solutions to the thin-sheet surface conductance `earth%tau`

Anna Kelbert with Claude

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Abstract

GLOBAL1D (`field1d.f90/field1d.s2.f90`) carries a single scalar parameter, `earth%tau` (units Siemens), representing an optional thin conducting sheet superimposed at the model's outer surface $r = r_0$; we denote it τ below. This note derives how τ enters the governing equations, identifies the natural dimensionless control parameter, and reports a numerical sensitivity test against the closed-form analytic solution for a uniformly conducting sphere. We find (i) no numerical advantage to a nonzero τ : the solver is exact and well-behaved at $\tau = 0$, which is also the case the closed-form reference itself assumes; (ii) the error introduced by a nonzero τ scales linearly with the dimensionless parameter $Q = \omega\mu_0\tau r_0$ over eight orders of magnitude, with the linear-regime coefficient implying the perturbation only reaches order unity near $Q \sim 40$; and (iii) a dedicated Earth-scale test (matching the conductor's own electrical size to the toy sphere's, to isolate the effect of Q alone) confirms this threshold and identifies the mechanism beyond it: once Q dominates the conductor's own pre-jump value, the boundary condition is controlled by the artificial thin sheet rather than the layered conductor, so the response *saturates* in magnitude and can *overshoot and reverse sign* in phase, rather than growing without bound. (iv) A further check using the actual layered Earth conductivity model (`LWS/layered.GDE_rho.prm`) at the production period of 3600 s shows that, at the same Q , the real Earth's much higher conductivity suppresses the perturbation well below what the matched-regime toy-sphere comparison of (iii) predicted — the $Q \sim 40$ threshold is specific to that toy conductor's own electrical regime, not a universal constant.

1 How τ enters the equations

Both solvers solve for the toroidal potential $T_l(r)$ via a transfer-matrix recursion through the layered conductivity structure, matched to an insulating atmosphere above r_0 . Ordinary continuity of the layered-earth solution requires both T and T' continuous across every interface. At the model's outer surface, *both* solvers instead apply a modified matching condition (`field1d.s2.f90` eq. 17; `field1d.f90`, algebraically identical, independently written):

$$T'_l(r_0^+) = T'_l(r_0^-) - i\omega\mu_0\tau T_l(r_0), \quad T_l(r_0^+) = T_l(r_0^-). \quad (1)$$

T itself remains continuous; only T' jumps, by a term proportional to τ .

Physical interpretation. This is the classical thin-sheet (Price 1949) approximation. A sheet of conductance $\tau = \int \sigma dz$ carries an induced surface current $K = \tau E_{\text{tangential}}$ (Ohm's law integrated through the sheet's negligible thickness). Since $H_\theta, H_\phi \propto T'/r$ and $E_\theta, E_\phi \propto T$, Ampère's law relates a surface current directly to a jump in tangential H — exactly (1). Setting $\tau = 0$ removes the sheet entirely and recovers ordinary continuity.

Dimensionless control parameter. Comparing the added term in (1) against the natural scale of T' itself ($T'/T \sim O(l)/r_0$) gives the classical thin-sheet parameter

$$Q = \omega \mu_0 \tau r_0, \quad (2)$$

which governs the size of the perturbation independent of the specific (r_0, ω) used in any particular test or production run.

2 Numerical test

We use the same single homogeneous-sphere configuration as the existing closed-form regression test (`test_unit_sphere.f90`: $r_0 = 1000$ m, $\sigma = 1$ S/m, $\omega = 1$ rad/s, unit external ($l=2, m=+1$) multipole source), for which S2 already matches the analytic solution *exactly* at $\tau = 0$ (ratio = $1+0i$ for all five field components). The $\tau = 0$ output therefore serves directly as the analytic reference; no closed-form values need to be re-derived here. $H_r, H_\theta, H_\phi, E_\theta, E_\phi$ are extracted at one fixed grid point in the vacuum region immediately above the sphere for every τ .

2.1 Magnitude error vs. τ

Table 1 sweeps τ from 1 S down to 0, reporting Q (eq. 2) and the maximum relative error across all five field components against the $\tau = 0$ reference.

τ [S]	$Q = \omega \mu_0 \tau r_0$	max relative error (5 components)
1×10^0	1.257×10^{-3}	3.058×10^{-5}
1×10^{-1}	1.257×10^{-4}	3.058×10^{-6}
1×10^{-2}	1.257×10^{-5}	3.058×10^{-7}
1×10^{-3}	1.257×10^{-6}	3.058×10^{-8}
1×10^{-4}	1.257×10^{-7}	3.058×10^{-9}
1×10^{-5}	1.257×10^{-8}	3.058×10^{-10}
1×10^{-6}	1.257×10^{-9}	3.058×10^{-11}
1×10^{-7}	1.257×10^{-10}	3.058×10^{-12}
1×10^{-8}	1.257×10^{-11}	3.058×10^{-13} (roundoff floor)
0	0	0 (exact, by construction)

Table 1: Magnitude error vs. τ , referenced to the $\tau = 0$ analytic solution, for the toy sphere ($r_0 = 1000$ m, $\sigma = 1$ S/m, $\omega = 1$ rad/s). Error scales linearly with τ (and hence with Q) across eight decades, until double-precision roundoff floors out below $\tau \sim 10^{-6}$. The fit is error $\approx 0.024 Q$.

2.2 Magnitude ratio and phase shift of H at specific τ values

Tables 2–4 report the magnitude ratio $|H(\tau)/H(0)|$ and phase shift $\arg H(\tau) - \arg H(0)$ (degrees) of each magnetic field component in turn, for $\tau = 10^{-4}, 1, 10^2, 2.5 \times 10^3$ S.

Tables 3 and 4 are numerically identical. This is not a coincidence: H_θ and H_ϕ both derive from the same radial factor $T'(r)$, whose complex ratio to the $\tau = 0$ case does not depend on angular position, so the two components (evaluated at the same radius, different angles) necessarily agree. H_r (Table 2), driven by $T(r)$ rather than $T'(r)$, follows an independent (smaller-magnitude, opposite-sign) trend.

τ [S]	Q	$ H_r/H_{r,0} $	$\Delta\phi_{H_r}$ [°]
10^{-4}	1.257×10^{-7}	1.000000	1.7×10^{-7}
1	1.257×10^{-3}	0.999998	1.75×10^{-3}
10^2	0.1257	0.999722	0.1743
2.5×10^3	3.142	0.964484	3.086

Table 2: H_r magnitude ratio and phase shift relative to $\tau = 0$, for the toy sphere ($r_0 = 1000$ m, $\sigma = 1$ S/m, $\omega = 1$ rad/s).

τ [S]	Q	$ H_\theta/H_{\theta,0} $	$\Delta\phi_{H_\theta}$ [°]
10^{-4}	1.257×10^{-7}	1.000000	-3×10^{-8}
1	1.257×10^{-3}	1.000000	-3.00×10^{-4}
10^2	0.1257	1.000051	-0.02992
2.5×10^3	3.142	1.006426	-0.5060

Table 3: H_θ magnitude ratio and phase shift relative to $\tau = 0$, for the toy sphere ($r_0 = 1000$ m, $\sigma = 1$ S/m, $\omega = 1$ rad/s).

3 Earth-scale test: is Q alone sufficient, and why does the response become nonlinear?

The toy sphere has $r_0 = 1000$ m, $\sigma = 1$ S/m, $\omega = 1$ rad/s (as in §2); the real Earth has $r_0 \approx 6.371 \times 10^6$ m (6371 $\times$ larger). Since $Q \propto r_0$ (eq. 2), the *same* τ produces a vastly larger Q at Earth scale — Table 5 evaluates this at the NA2026 project’s own periods (162 s, 3600 s).

Methodology: isolating Q from the conductor’s own regime. Naively re-running the toy sphere with $r_0 = 6.371 \times 10^6$ m but the same $\sigma = 1$ S/m, $\omega = 1$ rad/s would *also* push the conductor’s own electrical size $|k_l|r_0 = \sqrt{\omega\mu_0\sigma} r_0$ from ≈ 1.12 (order-1 induction, the toy sphere’s regime) to ≈ 1400 (extreme skin effect) — a second, unrelated regime change that would confound any observed nonlinearity: we would no longer know whether a deviation from linear-in- Q behavior is caused by τ growing large, or by the conductor itself behaving completely differently. To isolate the τ/Q effect cleanly, σ at each Earth-scale period is instead chosen so that $|k_l|r_0$ *equals* the toy sphere’s own value (1.121), holding the conductor’s regime fixed while only r_0 (and hence Q , for the same τ) changes. This gives $\sigma = 6.35 \times 10^{-7}$ S/m at 162 s and $\sigma = 1.41 \times 10^{-5}$ S/m at 3600 s — unrealistically resistive for an actual whole-Earth sphere, but that is expected and intentional: this is a controlled numerical experiment isolating one mechanism, not a production-accurate Earth model. $H_\theta = H_\phi$ again (same argument as §2.2), so only one column pair is shown for them.

Refining the linear-regime threshold. Table 1’s fit, error $\approx 0.024 Q$, implies the perturbation only reaches order unity once $Q \sim 1/0.024 \approx 42$ — not $Q \sim 1$ as the crude order-of-magnitude argument in §1 suggested. This is exactly consistent with Tables 6–7: at $Q = 0.31$ –1.4 the response is still closely linear in Q (e.g. $\Delta\phi_{H_r}$ at $Q = 0.31$ (162 s) and $Q = 1.4$ (3600 s) both fall close to the toy-sphere fit scaled to that Q), while by $Q = 31$ –35 (comparable to the refined $Q \sim 42$ threshold) the magnitude ratio has already dropped by $\sim 12\%$ — clearly nonlinear.

τ [S]	Q	$ H_\phi/H_{\phi,0} $	$\Delta\phi_{H_\phi}$ [°]
10^{-4}	1.257×10^{-7}	1.000000	-3×10^{-8}
1	1.257×10^{-3}	1.000000	-3.00×10^{-4}
10^2	0.1257	1.000051	-0.02992
2.5×10^3	3.142	1.006426	-0.5060

Table 4: H_ϕ magnitude ratio and phase shift relative to $\tau = 0$, for the toy sphere ($r_0 = 1000$ m, $\sigma = 1$ S/m, $\omega = 1$ rad/s).

τ [S]	$Q_{\text{Earth, 162 s}}$	$Q_{\text{Earth, 3600 s}}$
10^{-4}	3.1×10^{-5}	1.4×10^{-6}
1	0.31	0.014
10^2	31	1.4
2.5×10^3	776	35

Table 5: Q at Earth scale for the same τ values as Tables 2–4.

Why the response saturates rather than diverging. From eq. (1), with $T_l(r_0) = 1$ by construction, $T'_l(r_0^+) = t_{np}(l) - iQ$, where $t_{np}(l)$ is the conductor’s own (τ -independent) pre-jump value. Once $Q \gg |t_{np}(l)|$, this sum is dominated by the $-iQ$ term regardless of the conductor’s own properties: $T'_l(r_0^+) \rightarrow -iQ$, a fixed -90° phase relative to $T_l(r_0)$, independent of τ once τ is large enough. This is why the magnitude ratio *saturates* (Tables 6–7: essentially unchanged between $Q = 31$ and $Q = 776$, and between $Q = 1.4$ and $Q = 35$) rather than continuing to grow, and why the phase shift can *overshoot and reverse sign* (162 s: $+0.98^\circ$ at $Q = 31$ to -0.20° at $Q = 776$; 3600 s: $+1.76^\circ$ at $Q = 1.4$ down to $+0.85^\circ$ at $Q = 35$) rather than growing monotonically: for very large τ the physics is no longer governed by the layered conductor at all, but entirely by the artificial thin sheet, and further increasing τ has diminishing effect on the already- τ -dominated boundary condition. This is a genuine physical saturation, not a numerical artifact: the same qualitative behavior (rise, peak/plateau, mild reversal) appears at both Earth-scale periods, at Q values that differ by a factor of ~ 20 , using two independently matched conductor regimes.

4 Realistic layered Earth conductivity model

§3’s Earth-scale test used a homogeneous sphere with σ chosen only to match the toy sphere’s own $|k_l|r_0$ — a controlled experiment isolating the Q mechanism, not a realistic Earth. Here we repeat the same τ sweep at Earth radius and the NA2026 project’s 3600 s period, but with the *actual* layered conductivity profile used in production runs, `LWS/layered_GDE_rho.prm`, read the same way `FWD1D.f90` builds `earth%layer/` `earth%sigma` from a `layers`-format model file: 32 layers given as depth below the surface (km) and $\log_{10} \rho$ (ρ in $\Omega \cdot \text{m}$), converted to $\sigma = 10^{-\log_{10} \rho}$, plus a hardcoded $\sigma = 10^5$ S/m “core” layer beneath the deepest defined depth (3500 km). Table 8 quotes every depth/ σ pair used.

The perturbation is markedly smaller than the matched-regime toy comparison predicted. Q depends only on ω , τ , r_0 (eq. 2), not on conductivity, so Table 9 uses *exactly* the same Q values as Table 7. Yet at $Q = 34.9$ ($\tau = 2500$ S), the realistic model gives $|H_r/H_{r,0}| = 0.980$ and $\Delta\phi_{H_r} = 0.026^\circ$, versus 0.882 and 0.85° for the matched-regime homogeneous sphere — roughly $6\times$

τ [S]	Q	$ H_r/H_{r,0} $	$\Delta\phi_{H_r}$ [°]	$ H_{\theta,\phi}/H_{\theta,\phi,0} $	$\Delta\phi_{H_{\theta,\phi}}$ [°]
10^{-4}	3.11×10^{-5}	1.000000	4.3×10^{-5}	1.000000	-7.4×10^{-6}
1	0.3105	0.999065	0.4254	1.000173	-0.07295
10^2	31.05	0.882199	0.9808	1.020275	-0.13990
2.5×10^3	776.3	0.878969	-0.1977	1.020790	0.03518

Table 6: H magnitude ratio and phase shift relative to $\tau = 0$, Earth radius, period 162 s, $\sigma = 6.35 \times 10^{-7}$ S/m (matched $|k_l|r_0$).

τ [S]	Q	$ H_r/H_{r,0} $	$\Delta\phi_{H_r}$ [°]	$ H_{\theta,\phi}/H_{\theta,\phi,0} $	$\Delta\phi_{H_{\theta,\phi}}$ [°]
10^{-4}	1.40×10^{-6}	1.000000	1.9×10^{-6}	1.000000	-3.3×10^{-7}
1	0.01397	0.999976	0.01932	1.000004	-0.00332
10^2	1.397	0.989809	1.7626	1.001871	-0.29856
2.5×10^3	34.93	0.881538	0.8515	1.020380	-0.12055

Table 7: H magnitude ratio and phase shift relative to $\tau = 0$, Earth radius, period 3600 s, $\sigma = 1.41 \times 10^{-5}$ S/m (matched $|k_l|r_0$).

smaller in both magnitude deviation and phase shift at the identical Q . The mechanism from the previous paragraph explains why: saturation sets in once $Q \gg |t_{np}(l)|$, and $|t_{np}(l)|$ — the conductor’s own pre-jump value — is set by the conductivity structure, not by Q . The real Earth model is far more conductive through most of its depth (up to $\sigma \approx 1.7$ S/m in the mantle, $\sigma = 10^5$ S/m in the core; see Table 8) than the thin, $|k_l|r_0$ -matched homogeneous sphere used in §3 to isolate the Q effect, giving it a correspondingly larger $|t_{np}(l)|$ and therefore requiring a larger Q before the $-iQ$ term in the jump condition dominates. The phase shift is still non-monotonic in Q here too (0.058° at $Q = 1.4$, falling to 0.026° at $Q = 34.9$) — the same qualitative saturation/overshoot behavior identified in §3, just at a much smaller absolute scale and a lower onset Q than that section’s toy-conductor threshold. **Conclusion: the $Q \sim 40$ threshold derived in §2.1/§3 is specific to that toy conductor’s own electrical regime, not a universal property of Q alone** — a genuinely more conductive Earth model tolerates a larger τ (at fixed r_0, ω) before departing from linear response. This makes the practical recommendation below (set $\tau = 0$) even safer for production runs than §3 alone would suggest: at the current default $\tau = 10^{-4}$ S, $Q \approx 1.4 \times 10^{-6}$ at 3600 s regardless of which conductivity model is used, many orders of magnitude below where any version of this nonlinearity begins.

5 Discussion

No numerical advantage to nonzero τ . τ appears in exactly one place in each solver: the additive term in (1). It is never a divisor, matrix pivot, or regularizer, so $\tau = 0$ introduces no degeneracy — consistent with the exact match already obtained at $\tau = 0$ in the existing regression test. There is no floating-point or conditioning reason to prefer a small nonzero value over exactly zero.

Practical recommendation. For runs that are not deliberately modeling a real conductive surface sheet, set $\tau = 0$: it is the mathematically clean choice, matches the validated exact-match test case, and removes an otherwise-unexplained free parameter. FWD1D.f90’s current default

depth [km]	$\log_{10} \rho$	σ [S/m]	depth [km]	$\log_{10} \rho$	σ [S/m]
10	2.000000	1.000×10^{-2}	800	0.430060	3.715×10^{-1}
40	3.000000	1.000×10^{-3}	850	0.279000	5.260×10^{-1}
110	3.000000	1.000×10^{-3}	900	0.279000	5.260×10^{-1}
160	2.522879	3.000×10^{-3}	1000	-0.228100	1.691
210	2.154902	7.000×10^{-3}	1100	-0.228100	1.691
260	1.745627	1.796×10^{-2}	1200	-0.228100	1.691
310	1.667885	2.148×10^{-2}	1300	-0.228100	1.691
360	1.688480	2.049×10^{-2}	1400	-0.228100	1.691
410	1.709952	1.950×10^{-2}	1500	-0.228100	1.691
460	1.699400	1.998×10^{-2}	1620	-0.228100	1.691
510	1.645104	2.264×10^{-2}	1764	-0.228100	1.691
560	1.540801	2.879×10^{-2}	1936.8	-0.228100	1.691
610	1.379495	4.174×10^{-2}	2144.16	-0.228100	1.691
660	1.153394	7.024×10^{-2}	2400	-0.228100	1.691
710	0.869989	1.349×10^{-1}	2900	-1.000000	10.00
760	0.593626	2.549×10^{-1}	3500	-2.000000	100.0
			(core, $r < r_0 - 3500$ km)	—	1.000×10^5

Table 8: Layer depths and conductivities of the layered Earth model, `LWS/layered_GDE_rho.prm` (32 layers, $\sigma = 10^{-\log_{10} \rho}$, plus the hardcoded core layer), used for Table 9.

τ [S]	Q	$ H_r/H_{r,0} $	$\Delta\phi_{H_r}$ [°]	$ H_{\theta,\phi}/H_{\theta,\phi,0} $	$\Delta\phi_{H_{\theta,\phi}}$ [°]
10^{-4}	1.40×10^{-6}	1.000000	6×10^{-8}	1.000000	-1×10^{-8}
1	0.01397	0.999987	6.37×10^{-4}	1.000002	-9.56×10^{-5}
10^2	1.397	0.998695	0.05833	1.000201	-0.00872
2.5×10^3	34.93	0.979892	0.02557	1.003066	-0.00138

Table 9: H magnitude ratio and phase shift relative to $\tau = 0$, Earth radius, period 3600 s, *realistic* layered conductivity (Table 8) rather than the $|k_l|r_0$ -matched homogeneous sphere of Table 7. Same τ, Q values as every other table in this report.

$\tau = 10^{-4}$ has no numerical justification we could identify and can be replaced by 0 without any accuracy cost.

Scale dependence. Since $Q \propto r_0$ (eq. 2), the toy sphere’s Q values ($\tau = 2500$ S gives only $Q = 3.14$) do *not* directly represent Earth-scale behavior — at Earth’s radius the same τ values give Q up to 776 (Table 5), well beyond where Table 1’s linear fit holds. §3 evaluates this directly rather than extrapolating: at $\tau = 10^{-4}$ S, $Q_{\text{Earth}} \sim 10^{-5}$ – 10^{-6} , confirming the current default is negligible at Earth scale too and can safely be replaced by 0. At $\tau = 100$ – 2500 S, Q_{Earth} reaches the regime where §3’s toy-conductor comparison first showed nonlinearity ($Q \gtrsim 40$) — but §4 shows this specific threshold is conductor-dependent, not a property of Q alone: repeating the same τ sweep with the actual production conductivity model (`LWS/layered_GDE_rho.prm`) at 3600 s gives a perturbation roughly $6\times$ smaller, at the identical Q , than the matched-regime toy sphere predicted, because the real Earth’s higher conductivity gives it a correspondingly larger $|t_{np}(l)|$. This is not merely a mechanism demonstration but a production-representative estimate already obtained (§4,

Table 9): if a nonzero τ representing a genuine conductive surface feature (e.g. the ocean) is intentionally used at Earth scale, its quantitative effect on this specific conductivity model is now known directly, not merely bounded by the toy-sphere mechanism study.